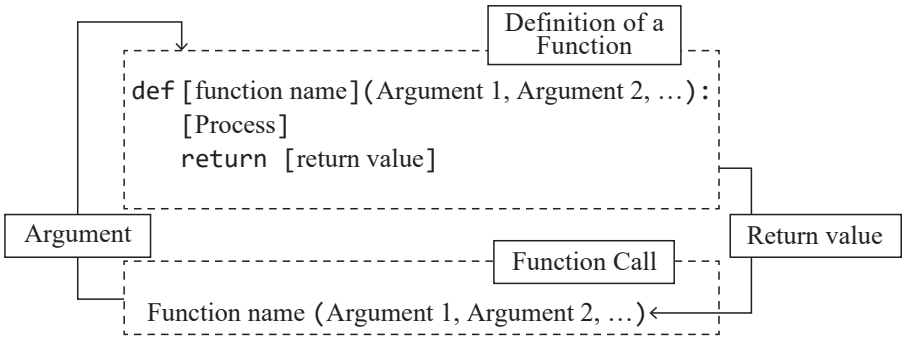
12-5 Application of Programming [2] (Python)

Point!

Functions

(1) ([1]**Function**) : A set of operations performed by a computer described as a cohesive unit. A variable specified to receive a value within a function is referred to as an ([2]**argument**). Using ([3]**return**) makes it possible to pass the value resulting from the process within the function to the caller as a ([4]**return value**). In addition, when a function is used, it is necessary to perform the process of ([5]**calling**) the function.



<Example> Example of a function with one argument: A program to find the function y = 2x

01	`def function(x):`	Defines a `function` called ([6]**function**).
02	`    y = 2 * x`	Assigns the value from `2*x` to variable "`y`".
03	`    return y`	Returns "`y`" as ([7]**return value**).
04	`print(function(5))`	Passes "`x`" as 5 to the `function`.

Execution result

```
10
```

<Example> Example of a function with two arguments: A program to find the area of a triangle

01	`def area(base, height):`	Defines a function called "`area`".
02	`    S= base*height/2`	Assigns the value from "`base*height/2`" to variable "`S`".
03	`    return S`	Returns "`S`" as the return value.
04	`print(area(10, 5))`	Passes "`base`" as ([8]`10`) and "`height`" as ([9]`5`) to the function "`area`".

Execution result

```
25
```

(2) Functions include ([10]**built-in functions**) such as "`print()`" that can be used without definition, and ([11]**user-defined functions**) that you can define.

Warm Up

Answer the following questions.

(1) In the following program, the function "`circle`" is used to calculate the area of a circle. Give the program that fits in blanks **A** and **B**. Also, give the value displayed when row [1] is executed.

```
 A  circle(r):
    S = r * r * 3.14
    B  S
a = circle(5)
print(a)      # [1]
```

(2) In the following program, the function "`judge`" takes the test score as an argument and displays "`Pass`" if the score is 80 or above, and "`Fail`" if it is below 80. Give the program that fits in blanks **A** and **B**.

```
def judge(score):
    if score  A  80:
        print('Pass')
    B :
        print('Fail')
judge(40)
```

Explanation

(1) In programming, the function is defined as shown below.

```
def function name(argument):
    [Process]
    return [return value]
```

Therefore, the programs that fit in blanks **A** and **B** are as follows.

A: `def`, **B**: `return`

In addition, when the function "`circle`" is called in [1], 5 is passed as argument "`r`" in the function "`circle`".

Therefore, when row [1] is executed, it is `5 × 5 × 3.14 = 78.5`, so `78.5` is displayed.

(2) In the programming, a conditional branch was defined as shown below.

```
if conditional expression:
    [Process when a condition is true]
else:
    [Process when a condition is false]
```

Therefore, the programs that fit in blanks **A** and **B** are as follows.

A: `>=`, **B**: `else`

Try

Answer the following questions.

(1) Choose the term that best fits into the blanks [1] and [2] from the options **A** to **D** below, and answer using the letters.

```
def function name ( [1] ):
    [Process]
    return [2]
```

A Argument **B** Return value **C** Complement **D** Random number

(2) In the following program, the function "`area`" is used to calculate the area of a triangle. Give the program that fits in blanks **A** and **B**. Also, give the values displayed when rows [1] and [2] are executed.

```
def area(base, height):
    S = A
    B S
a = area(10, 5)
b = area(6, 7)
print(a)      # [1]
print(b)      # [2]
```

(3) In the following program, the function "`celsius_to_fahrenheit`" converts temperatures from `Celsius` to `Fahrenheit`. Give the program that fits in blanks **A** and **B**.

```
def celsius_to_fahrenheit(celsius):
    fahrenheit = (celsius * 9/5) + 32
    return A
temp_celsius = 25
result = celsius_to_fahrenheit( B )
print(result)
```

(4) In the following program, test scores are given as arguments to the function "`evaluate`". In addition, the relationship between scores and grades is according to the table on the right. Give the program that fits in the blanks **A** to **C**.

Score	Grade
80 or higher	A
50 or higher but less than 80	B
Less than 50	C

```
def evaluate(score):
    if score A 80:
        print('A')
    B score C 50:
        print('B')
    else:
        print('C')
evaluate(40)
```

13–1 About Generative AI

Point!

Generative AI performs intellectual tasks on behalf of humans, so using it effectively can streamline your workflow.
In the real world, Generative AI is used in various situations, such as generating new ideas, summarizing texts, supporting studies, and programming.

Generative AI has learned from vast amounts of information worldwide. It generates natural text by selecting **"the word most likely to come next"** in response to a user's **"Prompt"** (instruction).

For example, if you provide the prompt **"What is the capital of Egypt?"**: The AI selects "Cairo" as the word most likely to follow "The capital of Egypt is," and generates the sentence "The capital of Egypt is Cairo."
The key point is that the Generative AI does not "know" as a fact that the capital of Egypt is Cairo; it is merely selecting the word with the highest probability of following the phrase "The capital of Egypt is."

Therefore, Generative AI does not always provide the correct answer. Due to its mechanism, it is **"certain to say incorrect things with some probability."** This is called **"Hallucination."**
When using Generative AI, please understand this characteristic and do not trust what the AI says 100%.

Warm Up

There are various Generative AIs in the world, such as ChatGPT, Google Gemini, and Microsoft Copilot. In this lesson, we will use Copilot.

First, let's create a text. Try giving Copilot the following prompt:
"Write a story about a cat wizard going on an adventure."

Next, try giving the following prompt:
"Rewrite it to include a dog companion."

In this way, simply by speaking to Generative AI in everyday language, you can create and modify text.

Next, try giving the following prompt:
"Give me a quiz about Egypt. Wait until I answer."

Generative AI can also generate questions using learned information. You can freely change the format of the questions, such as instructing it to wait for the user's answer, or changing from multiple-choice to open-ended questions.
By giving appropriate prompts to Generative AI, you can make use of it for learning.